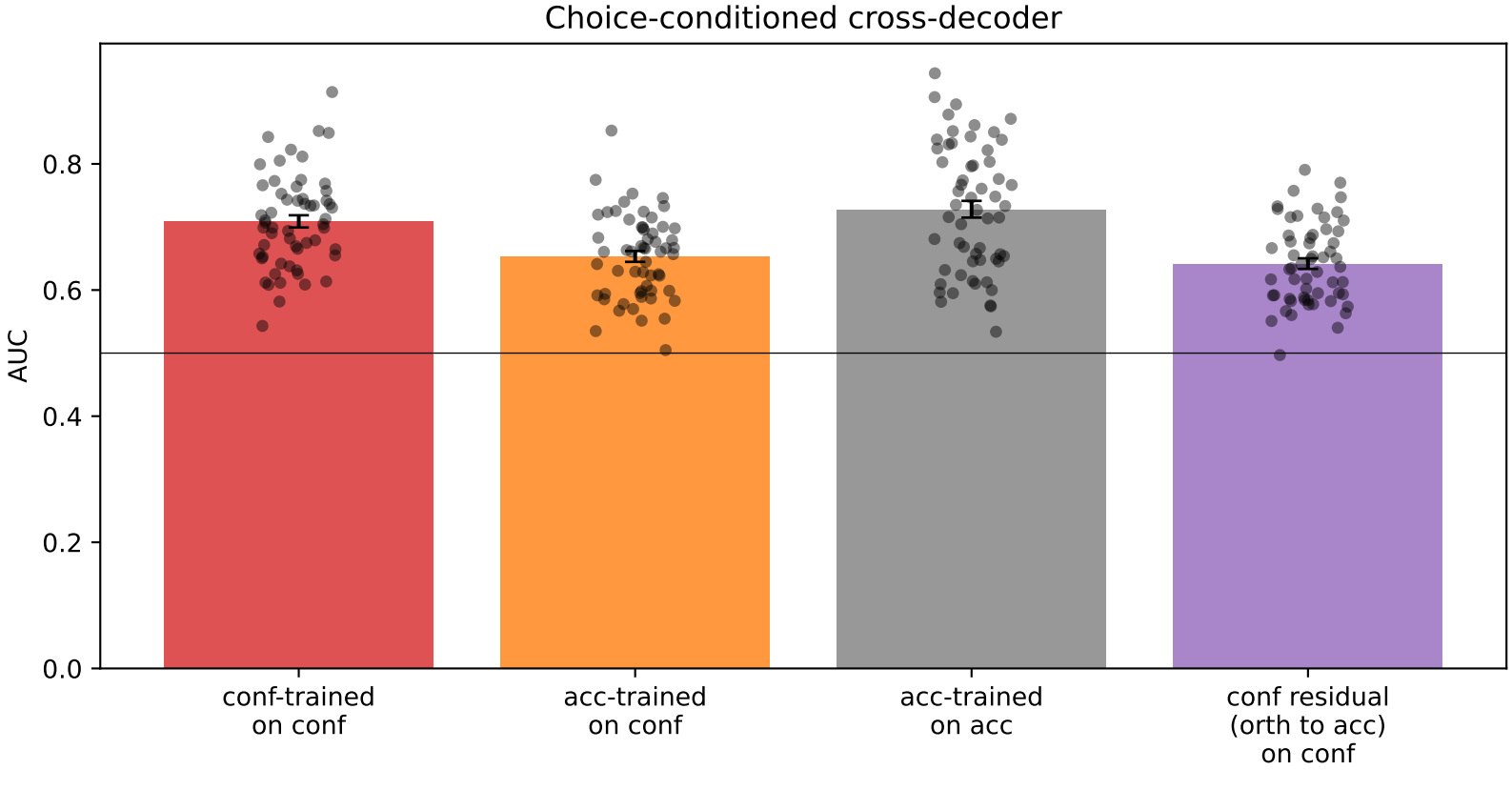
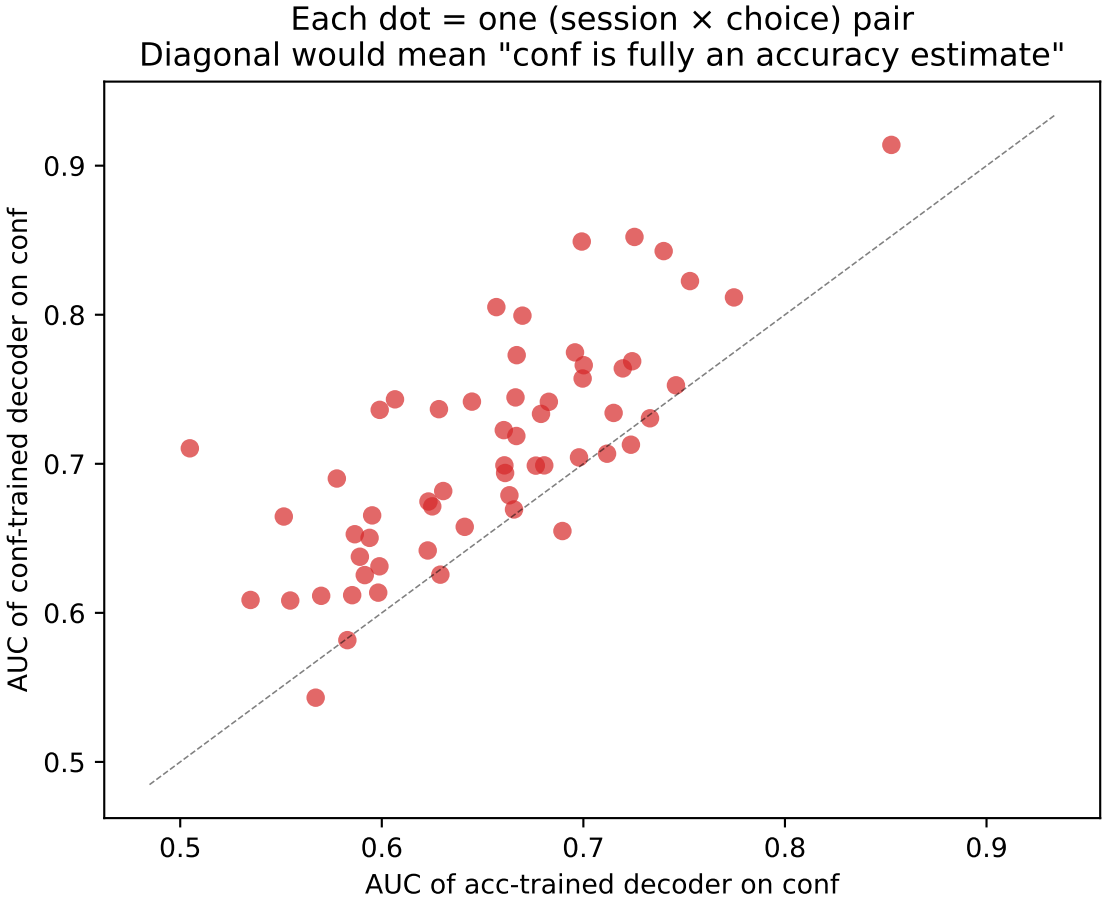
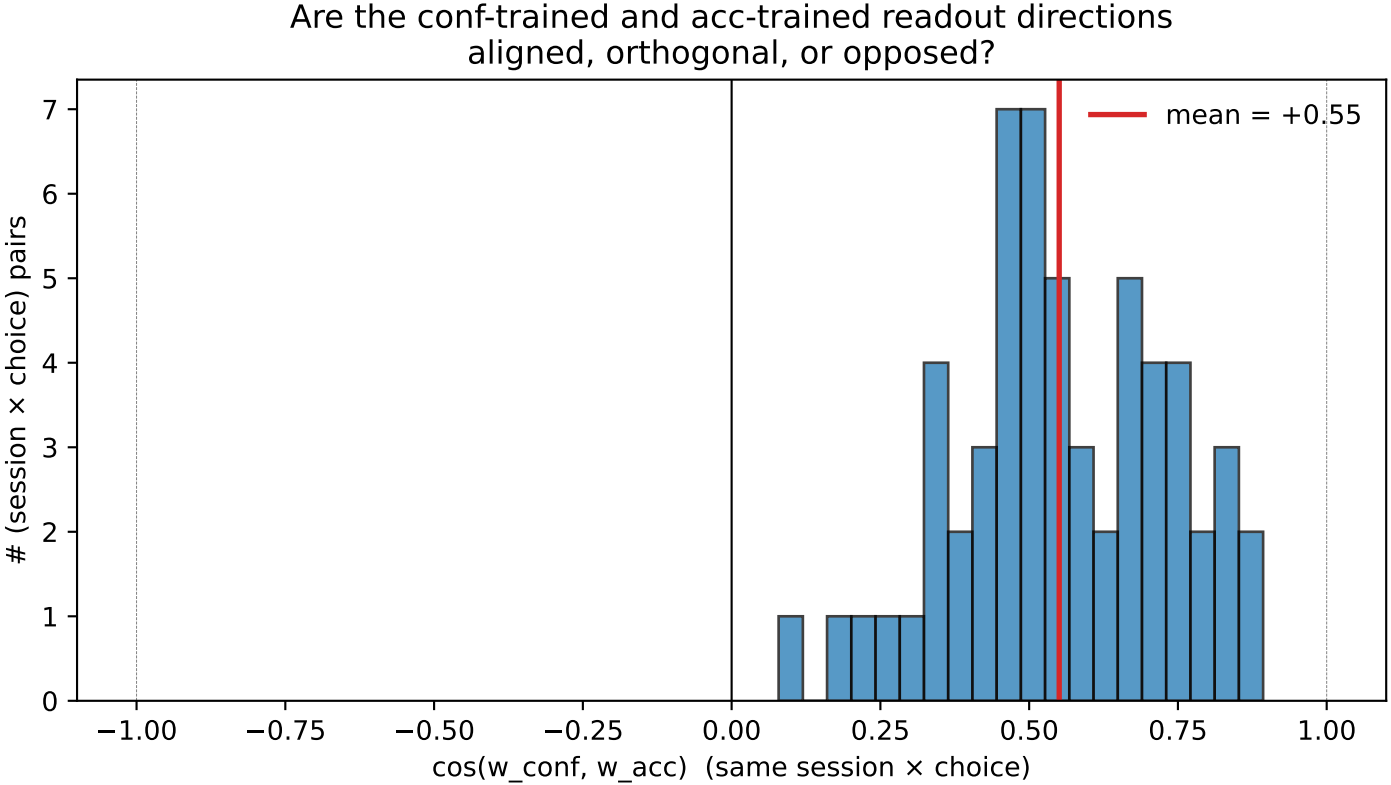

Choice-conditioned cross-decoder
AUC
0.0
0.2
0.4
0.6
0.8
conf-trained
on conf
acc-trained
on conf
acc-trained
on acc
conf residual
(orth to acc)
on conf


Each dot = one (session × choice) pair
Diagonal would mean "conf is fully an accuracy estimate"
AUC of conf-trained decoder on conf
AUC of acc-trained decoder on conf
0.5
0.6
0.7
0.8
0.9


Are the conf-trained and acc-trained readout directions
aligned, orthogonal, or opposed?
mean = +0.55
# (session × choice) pairs
0
1
2
3
4
5
6
7
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
cos(w_conf, w_acc) (same session × choice)

```
Choice-conditioned cross-decoder (29 sessions x 2 choices = 58 pairs)

AUC (mean across pairs):
  conf-trained, scored on conf  = 0.709
  acc-trained,  scored on conf  = 0.653
  acc-trained,  scored on acc   = 0.728
  residual (conf orth to acc),  = 0.642
    scored on conf

Δ AUC (conf-trained − acc-trained, scored on conf):
  mean = +0.0556,  Wilcoxon p = 6.02e-10

cos(w_conf, w_acc) across 58 pairs:
  mean = +0.550,  median = +0.537

Interpretation:
  The two decoders are trained on the SAME neural activity, the SAME
  trials, with the SAME architecture (linear logistic).  The only
  difference is the *training label* (conf vs correct).
  Any AUC gap therefore reflects a genuine difference in the
  population codes for confidence vs correctness — the brain is
  reading a different linear combination of V1 activity when it
  reports confidence than would be required to predict correctness.
  The cosine similarity above quantifies this directly: it is the
  angle between the conf-trained and acc-trained readout weight
  vectors in neuron space (1 = identical direction, 0 = orthogonal,
  -1 = opposite).
```